

Tuple

Properties

```
In [ ]: 1)Immutable >> We can not change items.
        2)It is in Ordered.
        3)Duplicates Items are allowed.
        4)Tuple are enclosed by round brackets.
        5)Tuple items are comma seperated.
        6)Hetrogeneous Data Types:
           int ,float,complex,list,
           tuple, dict,pandas Series,Pandas DataFrame.
```

```
In [5]: tuple1=(3,4,5,6,7,8,9,4)
        tuple1
```

```
Out[5]: (3, 4, 5, 6, 7, 8, 9, 4)
```

```
In [2]: type(tuple1)
```

```
Out[2]: tuple
```

```
In [6]: set1={3,4,5,6,7,8,1,2,3}
        set1
```

```
Out[6]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [9]: dict1={'a':4,'b':300,'a':45}
        dict1
```

```
Out[9]: {'a': 45, 'b': 300}
```

```
In [12]: tuple1=(2,3,3,3,3,3,4,5,6,'python')
        tuple1
```

```
Out[12]: (2, 3, 3, 3, 3, 3, 4, 5, 6, 'python')
```

```
In [14]: tuple1=(2,3,3,3,3,3,4,5,6,'python')
        tuple1[0]=200
        tuple1
```

TypeError

Traceback (most recent call last)

Cell **In[14]**, line 2

```
1 tuple1=(2,3,3,3,3,3,4,5,6,'python')
----> 2 tuple1[0]=200
      3 tuple1
```

TypeError: 'tuple' object does not support item assignment

```
In [15]: x=[2,3,4,5,6,7,'python']
x[0]=200
x
```

```
Out[15]: [200, 3, 4, 5, 6, 7, 'python']
```

Indexing and slicing

```
In [16]: tuple1=(2,3,3,3,3,4,5,6,7,91,'python')
tuple1
```

```
Out[16]: (2, 3, 3, 3, 3, 4, 5, 6, 7, 91, 'python')
```

```
In [17]: tuple1[0]
```

```
Out[17]: 2
```

```
In [18]: tuple1[1]
```

```
Out[18]: 3
```

```
tuple1= (2, 3, 40, 5.56, 8.91, 'python')
+ve   = 0  1  2  3  4  5
-ve   =-6 -5 -4 -3 -2 -1
```

```
In [20]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
print(tuple1[3])
print(tuple1[-3])
```

```
5.56
5.56
```

```
In [21]: print(tuple1[2])
print(tuple1[-4])
```

```
40
40
```

Slicing:

```
In [ ]: tuple1[start_index:end_index:step_size]
```

```
In [ ]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
+ve   = 0  1  2  3  4  5
-ve   =-6 -5 -4 -3 -2 -1
```

```
In [22]: print(tuple1[1:4])
print(tuple1[-5:-2])
print(tuple1[1:-2])
print(tuple1[-5:4])
```

```
(3, 40, 5.56)
(3, 40, 5.56)
(3, 40, 5.56)
(3, 40, 5.56)
```

```
In [23]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
tuple1[::-1]
```

```
Out[23]: ('python', 8.91, 5.56, 40, 3, 2)
```

```
In [24]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
tuple1[5:1:-1]
```

```
Out[24]: ('python', 8.91, 5.56, 40)
```

```
In [25]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
tuple1[2:5:-1]# -1 means reverse start_index must be greater than end_index
```

```
Out[25]: ()
```

```
In [26]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
for i in tuple1:
    print(f"i=={i}")
```

```
i==2
i==3
i==40
i==5.56
i==8.91
i==python
```

```
In [27]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
for i,val in enumerate(tuple1):
    print(f"i=={i}")
```

```
i==0
i==1
i==2
i==3
i==4
i==5
```

```
In [28]: tuple1= (2, 3, 40, 5.56, 8.91, 'python')
         for i,val in enumerate(tuple1):
             print(f"Value at index {i} is {val}")
```

```
Value at index 0 is 2
Value at index 1 is 3
Value at index 2 is 40
Value at index 3 is 5.56
Value at index 4 is 8.91
Value at index 5 is python
```

How to create Tuple

```
In [29]: x=(3,4,5,6,7,8)
         x
```

```
Out[29]: (3, 4, 5, 6, 7, 8)
```

```
In [31]: x=()
```

```
In [32]: a=6
         b=8

         x=a,b
         print(x)
```

```
(6, 8)
```

```
In [33]: type(x)
```

```
Out[33]: tuple
```

```
In [34]: t1=(35,45)
         type(t1)
```

```
Out[34]: tuple
```

```
In [35]: t2=(35)
         type(t2)
```

```
Out[35]: int
```

```
In [36]: t2=(35,)
         type(t2)
```

```
Out[36]: tuple
```

```
In [37]: x=('data')
x
```

Out[37]: 'data'

```
In [39]: x=(45.7)
x
```

Out[39]: 45.7

```
In [40]: a=()
type(a)
```

Out[40]: tuple

Python Type Casting

```
In [42]: list1=[4,5,6,7,8,9]
tuple1=tuple(list1)
tuple1
```

Out[42]: (4, 5, 6, 7, 8, 9)

```
In [43]: x=(4,5,6,7,8,9)
y=list(x)

print(x)
print(y)
```

```
(4, 5, 6, 7, 8, 9)
[4, 5, 6, 7, 8, 9]
```

```
In [44]: x=(3,4,5,6,7,8,9)
x[2]=600
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[44], line 2
      1 x=(3,4,5,6,7,8,9)
----> 2 x[2]=600
```

TypeError: 'tuple' object does not support item assignment

Imp

```
In [45]: x=(3,4,5,6,7,8,9)
y=list(x)
y[2]=600
print(y)
x=tuple(y)
print(x)
```

```
[3, 4, 600, 6, 7, 8, 9]
(3, 4, 600, 6, 7, 8, 9)
```

```
In [46]: x=(4,5,6,7,8)
y=list(x)
y.insert(3,200)
x=tuple(y)
print(x)
```

```
(4, 5, 6, 200, 7, 8)
```

```
In [47]: tuple1=(3,4,5,6,7)
tuple2=(10,20,30,40,50)
tuple3=tuple1+tuple2
print(tuple1)
print(tuple2)
print(tuple3)
```

```
(3, 4, 5, 6, 7)
(10, 20, 30, 40, 50)
(3, 4, 5, 6, 7, 10, 20, 30, 40, 50)
```

Tuple Functions

1)index()

```
In [ ]: Tuple.index(value)
```

```
In [48]: tuple1=(10,20,30,40,50)
i=tuple1.index(40)
print(i)
```

```
3
```

```
In [49]: tuple1=(10,20,30,40,50)
i=tuple1.index(10)
print(i)
```

```
0
```

```
In [50]: tuple1=(10,20,30,40,50,20,20,20)
         i=tuple1.index(20)
         print(i)
```

1

```
In [51]: tuple1=(10,20,30,40,50,20,20,20,20,50)
         i=tuple1.index(2000)
         print(i)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[51], line 2
      1 tuple1=(10,20,30,40,50,20,20,20,20,50)
----> 2 i=tuple1.index(2000)
      3 print(i)

ValueError: tuple.index(x): x not in tuple
```

```
In [53]: tuple1=(10,20,30,40,50,20,20,20,20,50)
         index_list=[]
         for i,val in enumerate(tuple1):
             if val==20:
                 index_list.append(i)
         index_list
```

Out[53]: [1, 5, 6, 7, 8]

```
In [54]: tuple1=(10,20,30,40,50,20,20,20,20,50)
         index_list=[i for i,val in enumerate(tuple1) if val==20]
         index_list
```

Out[54]: [1, 5, 6, 7, 8]

```
In [55]: tuple1=(10,20,30,40,50,20,20,20,20,30)
         num=30
         index_list=[i for i,val in enumerate(tuple1) if val==num]
         index_list
```

Out[55]: [2, 9]

```
In [57]: tuple1=(10,20,30,40,50,20,20,20,20,30)
         num=300
         index_list=tuple([i for i,val in enumerate(tuple1) if val==num])
         index_list
```

Out[57]: ()

2)count()

```
In [ ]: The count() function counts the occurrences of the particular given value.
```

```
In [ ]: Tuple.count(Value)
```

```
In [58]: tuple1=(3,4,5,6,2,1,2,2,6,7,8,2,3)
tuple1.count(3)
```

```
Out[58]: 2
```

```
In [59]: tuple1=(3,4,5,6,2,1,2,2,6,7,8,2,3)
tuple1.count(2)
```

```
Out[59]: 4
```

Built in Function

sorted

```
In [60]: list1=[3,4,5,6,7]
sorted_list=sorted(list1)
sorted_list
```

```
Out[60]: [3, 4, 5, 6, 7]
```

```
In [61]: tuple1=(3,1,4,5,6,8)
sorted_tuple=sorted(tuple1)
print(tuple1)
print(sorted_tuple)
```

```
(3, 1, 4, 5, 6, 8)
[1, 3, 4, 5, 6, 8]
```

```
In [62]: tuple1=(3,1,4,5,6,8)
sorted_tuple=tuple(sorted(tuple1))
print(tuple1)
print(sorted_tuple)
```

```
(3, 1, 4, 5, 6, 8)
(1, 3, 4, 5, 6, 8)
```

reversed()

```
In [63]: tuple1=('math','Phy','che','eng')
rev_tuple=tuple(reversed(tuple1))

print(tuple1)
print(rev_tuple)
```

```
('math', 'Phy', 'che', 'eng')
('eng', 'che', 'Phy', 'math')
```

min()

```
In [64]: tuple1=(3,4,5,6,8)
         min(tuple1)
```

Out[64]: 3

max()

```
In [66]: tuple1=(9,3,45,62,1)
         max(tuple1)
```

Out[66]: 62

sum()

```
In [67]: tuple1=(3,4,5,6,8)
         sum(tuple1)
```

Out[67]: 26

```
In [68]: tuple1=(3,4,6,[2,3,4],9,11)
         len(tuple1)
```

Out[68]: 6

```
In [72]: tuple1=(3,4,6,[2,3,4],9,11)
         tuple1[3][0]=400
         tuple1
```

Out[72]: (3, 4, 6, [400, 3, 4], 9, 11)

```
In [73]: tuple1=(3,4,6,[2,3,4],9,11)
         tuple1[3].append(400)
         tuple1
```

Out[73]: (3, 4, 6, [2, 3, 4, 400], 9, 11)

```
In [ ]:
```